

Generative AI Quiz (Q&A Format)

1. What is Generative AI?

Answer: Generative AI is a type of artificial intelligence that creates new content such as text, images, code, audio, or videos based on learned patterns from training data.

2. What does GPT stand for?

Answer: GPT stands for *Generative Pre-trained Transformer*.

3. What is a Large Language Model (LLM)?

Answer: A Large Language Model is an AI model trained on massive text datasets to understand, generate, and process human language.

4. Which architecture is used in GPT models?

Answer: Transformer architecture.

5. What is a prompt in Generative AI?

Answer: A prompt is the input or instruction given to a model to generate a response.

6. What is prompt engineering?

Answer: Prompt engineering is the practice of designing and optimizing prompts to get accurate and relevant outputs from AI models.

7. What is hallucination in Generative AI?

Answer: Hallucination is when an AI model generates incorrect or fabricated information that appears believable.

8. What is fine-tuning?

Answer: Fine-tuning is the process of training a pre-trained model on a smaller, domain-specific dataset to improve performance for a specific task.

9. What is temperature in text generation?

Answer: Temperature controls randomness in output; lower values make responses more deterministic, higher values make them more creative.

10. What are tokens in LLMs?

Answer: Tokens are small units of text (words, subwords, or characters) that the model processes.

11. What is a Transformer in AI?

Answer: A Transformer is a deep learning model that uses attention mechanisms to process and generate sequences efficiently.

12. What is self-attention?

Answer: Self-attention allows a model to focus on relevant parts of input data when generating output.

13. What is RLHF?

Answer: Reinforcement Learning with Human Feedback is a method where human feedback is used to improve model responses.

14. What are diffusion models?

Answer: Diffusion models generate data (like images) by gradually removing noise from random data.

15. What is multimodal AI?

Answer: Multimodal AI can process and generate multiple types of data such as text, images, and audio.

16. What is zero-shot learning?

Answer: Zero-shot learning is when a model performs a task without prior examples.

17. What is few-shot learning?

Answer: Few-shot learning is when a model learns from a small number of examples.

18. What are embeddings?

Answer: Embeddings are numerical representations of data used for similarity and semantic understanding.

19. What is a vector database?

Answer: A vector database stores embeddings and enables fast similarity searches.

20. What is latency in AI systems?

Answer: Latency is the time taken for a model to generate a response.

 DevOps / AWS Scenario Q&A**21. How can Generative AI help in CI/CD pipelines?**

Answer: It can auto-generate pipeline scripts, debug errors, and optimize deployment workflows.

22. How is Generative AI used in log analysis?

Answer: It summarizes logs, detects anomalies, and identifies root causes.

23. How can Generative AI improve incident management?

Answer: It suggests fixes, automates troubleshooting, and reduces response time.

24. How does Generative AI help in Infrastructure as Code (IaC)?

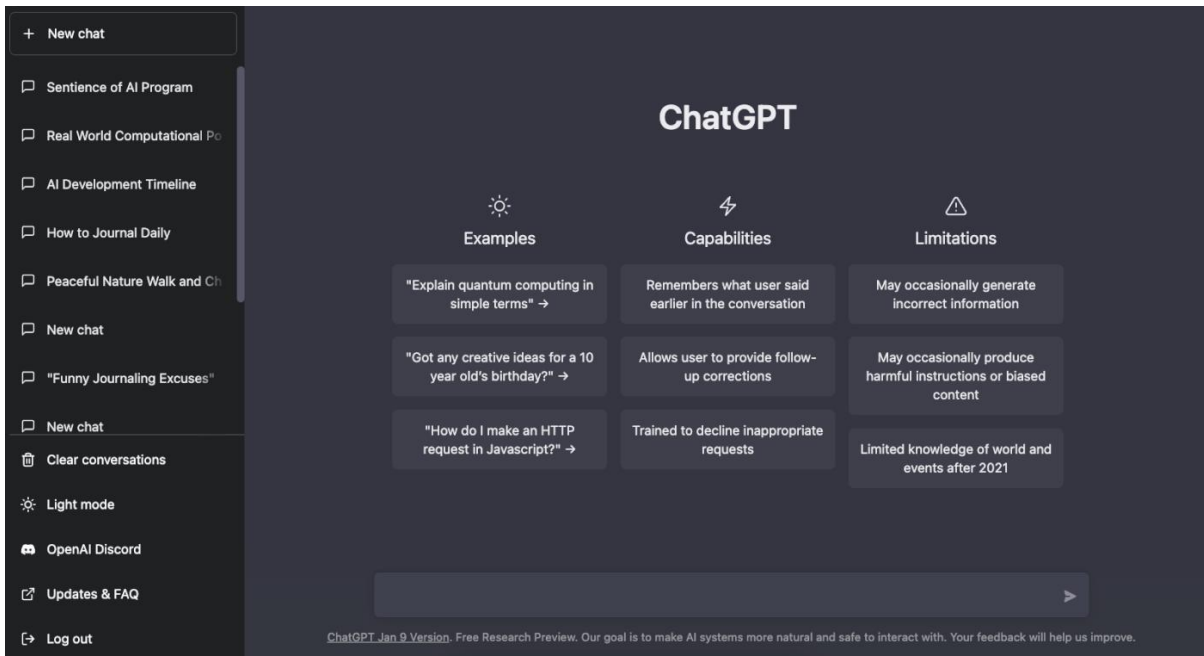
Answer: It can generate Terraform or CloudFormation templates automatically.

25. How can Generative AI reduce MTTR?

Answer: By quickly analyzing issues, suggesting solutions, and automating responses.

Examples of Generative AI Tools

Text Generation Tools



The screenshot displays the ChatGPT web interface. On the left is a sidebar with a list of chat sessions and navigation options. The main area features the 'ChatGPT' title and three columns: 'Examples', 'Capabilities', and 'Limitations'. Each column contains three items. At the bottom, there is a text input field and a footer note about the 'ChatGPT Jan 9 Version'.

Examples	Capabilities	Limitations
"Explain quantum computing in simple terms" →	Remembers what user said earlier in the conversation	May occasionally generate incorrect information
"Got any creative ideas for a 10 year old's birthday?" →	Allows user to provide follow-up corrections	May occasionally produce harmful instructions or biased content
"How do I make an HTTP request in Javascript?" →	Trained to decline inappropriate requests	Limited knowledge of world and events after 2021

ChatGPT Jan 9 Version. Free Research Preview. Our goal is to make AI systems more natural and safe to interact with. Your feedback will help us improve.





Hey there, I'm Claude.

I'm your AI assistant for working, imagining, and deep thinking.

Here's a few things you should know about me:



Ask me anything

Chat with me about anything from simple asks to complex ideas! Guardrails keep our chat safe.



Ad-free chats

I won't show you ads. My focus is being genuinely helpful to you.



You can improve Claude for everyone

We use your chats and coding sessions to train and improve Claude. You can change this setting anytime in your [Privacy Settings](#).

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- **ChatGPT** – Generates text, code, explanations, and answers questions
- **Google Gemini** – Multimodal AI for text, reasoning, and coding
- **Claude** – Good for long documents and safe responses
- **LLaMA** – Open-source LLM for research and apps



Use cases:

- Writing emails, blogs, documentation
- Code generation (DevOps scripts, YAML, Terraform)
- Chatbots & virtual assistants

Image Generation Tools

TEXT PROMPT

an armchair in the shape of an avocado [...]

AI-GENERATED IMAGES



[Edit prompt or view more images ↕](#)

TEXT PROMPT

a store front that has the word 'openai' written on it [...]

AI-GENERATED IMAGES



[Edit prompt or view more images ↕](#)





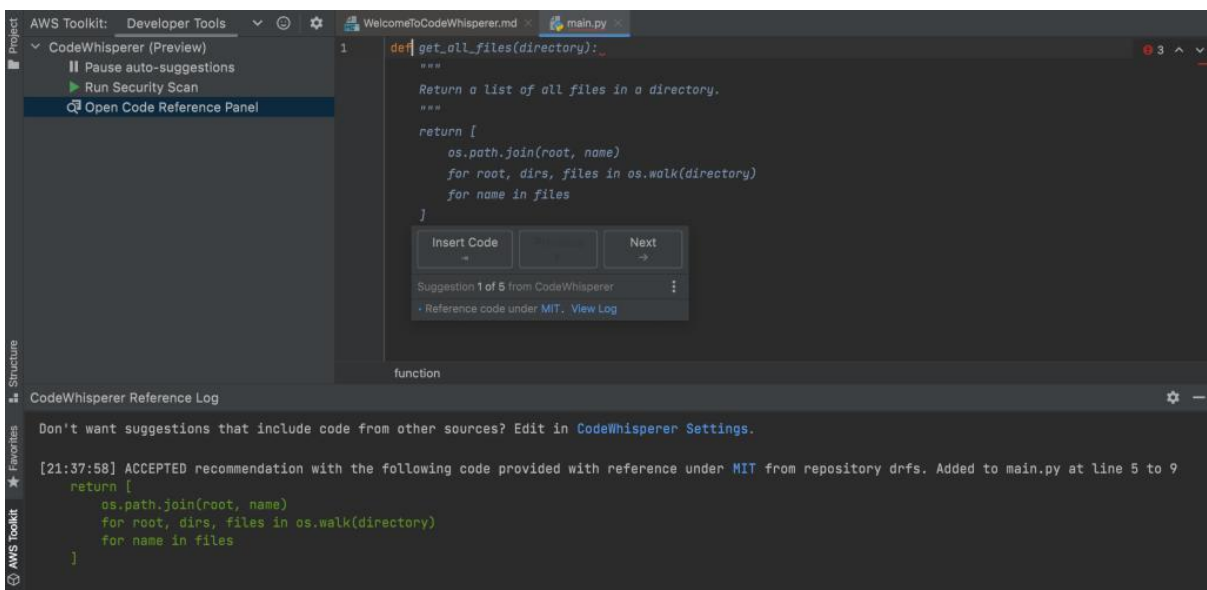
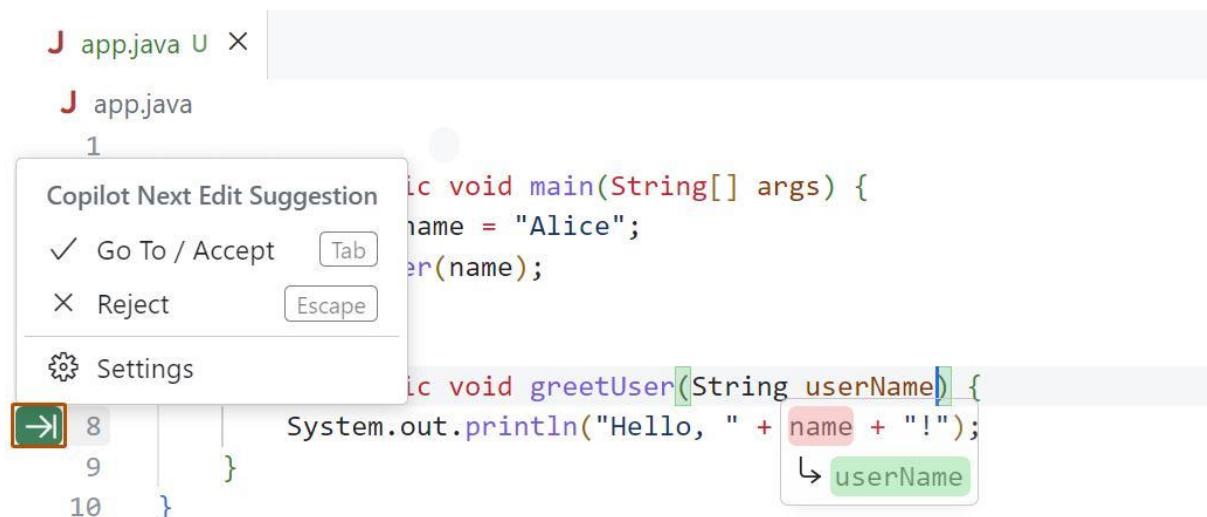
4

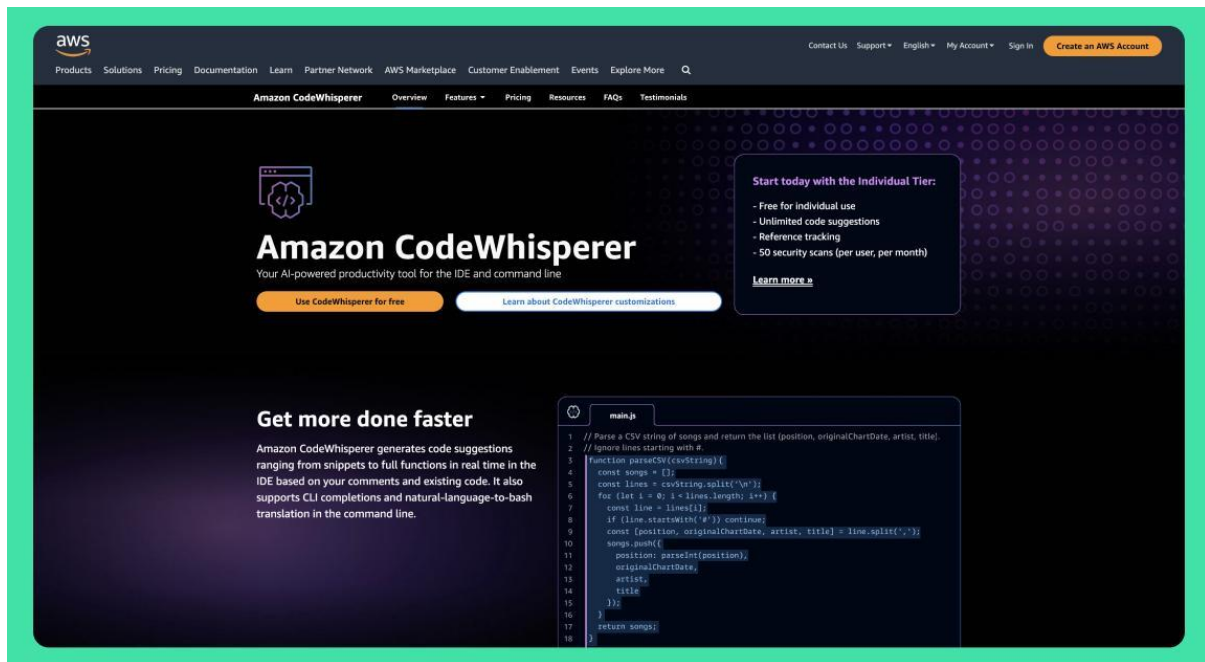
- **DALL·E** – Creates images from text prompts
- **MidJourney** – High-quality artistic images
- **Stable Diffusion** – Open-source image generation model

✓ **Use cases:**

- UI/UX design mockups
- Marketing creatives
- Architecture & concept art

📁 Code Generation Tools





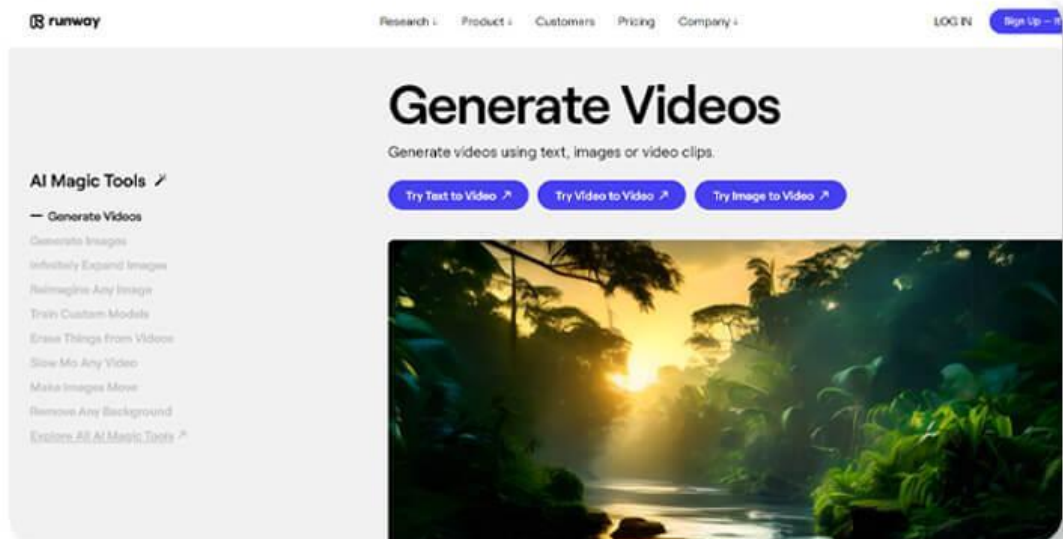
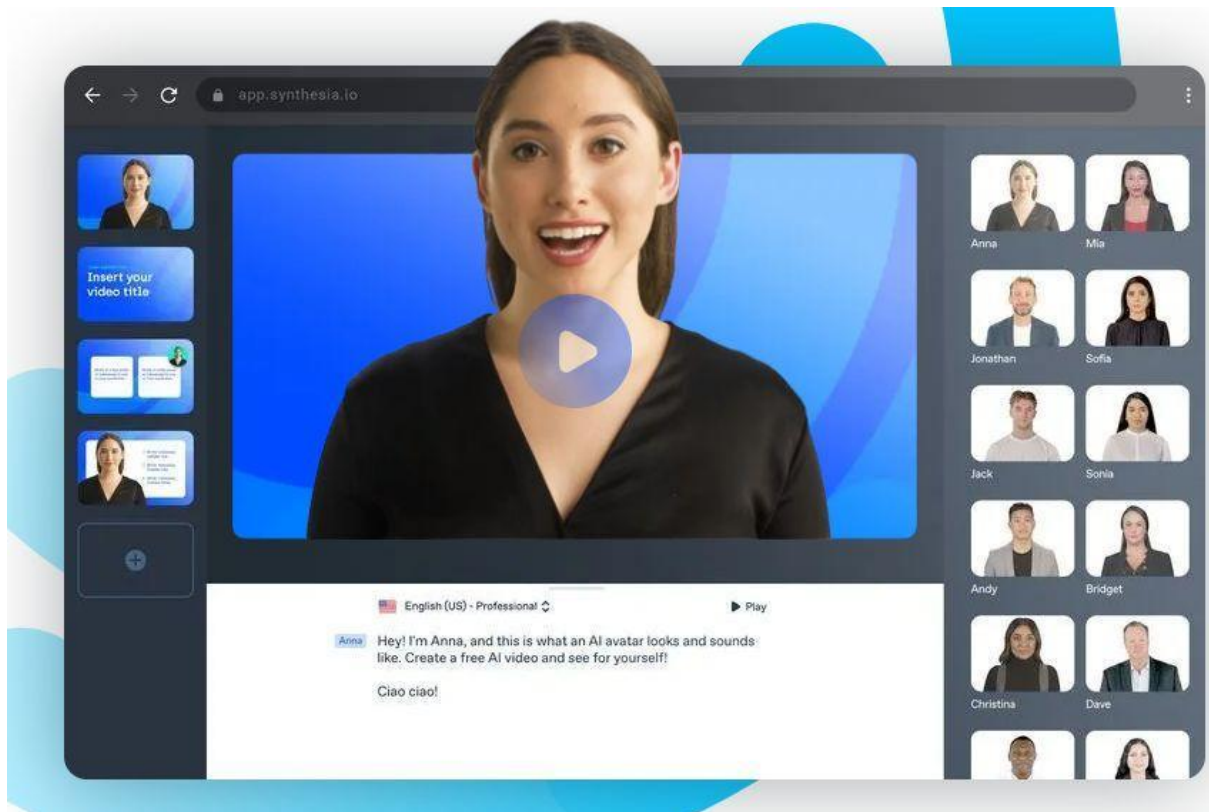
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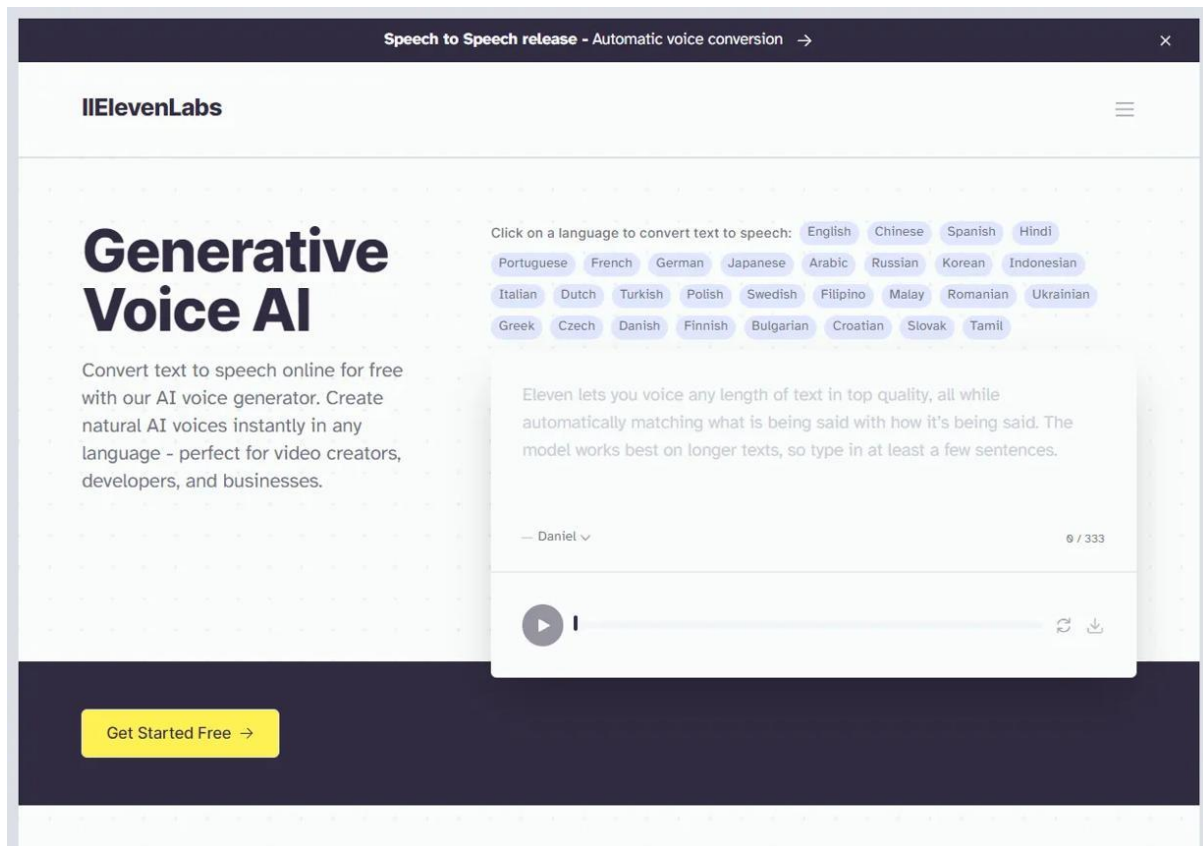
- **GitHub Copilot** – Suggests code in real-time
- **Amazon CodeWhisperer** – Helps with AWS-focused coding
- **Tabnine** – AI-powered code completion

✓ Use cases:

- Writing scripts (Bash, Python, YAML)
- Debugging code
- Auto-generating infrastructure templates

Video & Audio Generation Tools





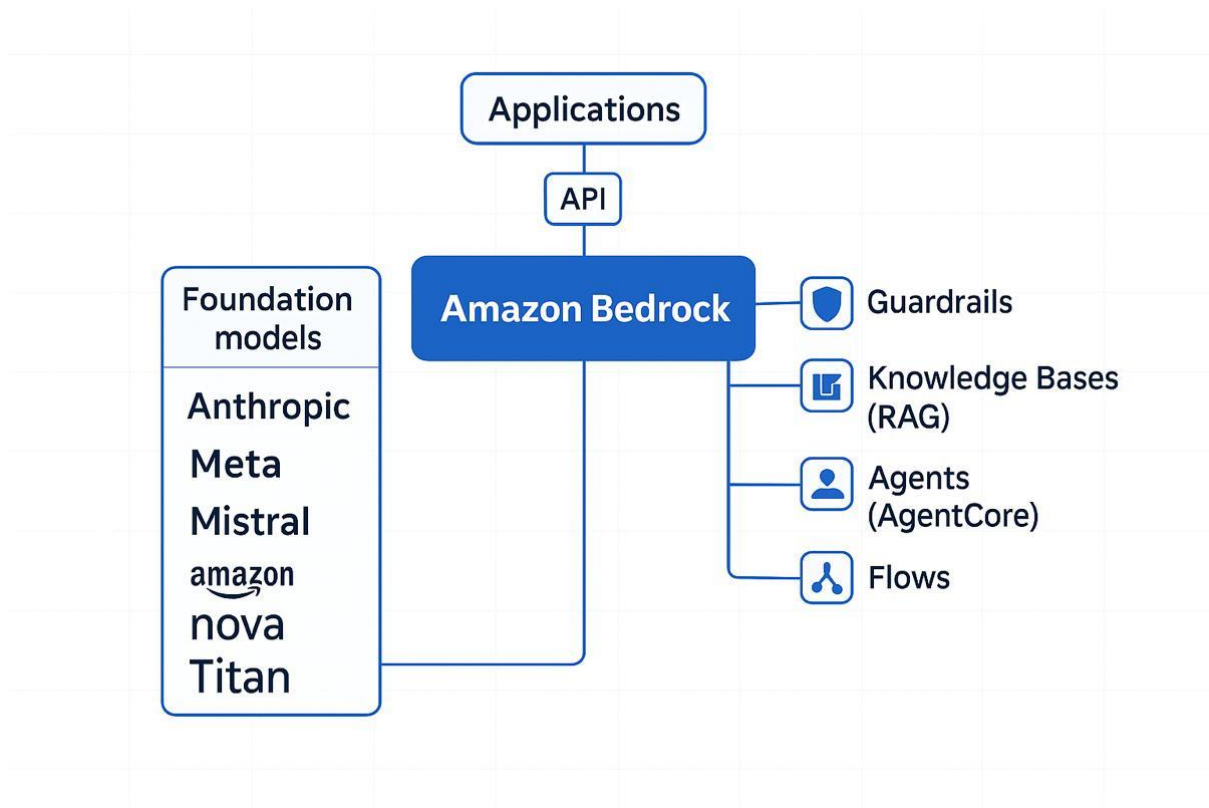
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- **Synthesia** – Creates videos with AI avatars
- **Runway ML** – Video generation & editing
- **ElevenLabs** – Realistic text-to-speech

✓ **Use cases:**

- Training videos
- YouTube content
- Voice assistants

 DevOps / AWS Specific Generative AI Tools



← ↻ <https://oai.azure.com/portal/8668f0ba655f4173af074199fcb4cbeb?tenantid=24a17522-d586-4f94-b056-e3781982833c>

Azure AI | Azure AI Studio

<< Azure AI Studio

Welcome to Azure OpenAI service

Explore the generative AI models, craft unique prompts for your use cases, and fine-tune select models.

Get started

Text generation



Chat playground

Design a customized AI assistant using ChatGPT. Experiment with GPT-3.5-Turbo and GPT-4 models.

[Try it now](#)

Text generation



Completions playground

Experiment with completions models for use cases such as summarization, content generation, and classification.

[Try it now](#)

Text generation



Bring your own data

PREVIEW

Connect and ground your data. Deploy to a web app or Power Virtual Agent bot (coming soon).

[Try it now](#)

Vertex AI Studio

Test, tune and deploy enterprise-ready generative AI

[TRY A TUTORIAL](#)

[DOCUMENTATION](#)

[API REFERENCE](#)

Generate with Gemini

OPEN FREEFORM

OPEN CHAT

Translate text →

Generate images →

Convert or synthesize speech →



Prompt management

Saved prompts and tools to make them better.



Tuning

Customize a model to your task

4

- **Amazon Bedrock** – Build GenAI apps using foundation models
- **Azure OpenAI Service** – Enterprise-grade GPT integration
- **Google Vertex AI** – Train & deploy AI models



Use cases:

- AI-powered monitoring & alerting
- Auto-generating CI/CD pipelines
- Intelligent log analysis



Quick Summary

Category	Tools
Text	ChatGPT, Gemini, Claude
Image	DALL·E, MidJourney, Stable Diffusion
Code	Copilot, CodeWhisperer, Tabnine
Video/Audio	Synthesia, Runway, ElevenLabs
DevOps AI	Bedrock, Azure OpenAI, Vertex AI